

FedMKS: A Hybrid Model and Knowledge Sharing Approach for Robust Federated Learning in Heterogeneous Networks

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Abstract—Federated Learning (FL) enables distributed clients to train a shared model without exchanging raw data, making it attractive for privacy-sensitive edge applications. Real deployments, however, face *system heterogeneity*: devices differ widely in compute capability, so no single model architecture fits all clients. Knowledge-distillation-based FL sidesteps this constraint by exchanging soft labels on a shared public set instead of weights, allowing each client to keep its own architecture. Yet under *statistical heterogeneity* (non-IID label skew), neither channel is uniformly reliable: weight averaging can collapse under client drift, while soft-label-only distillation degrades when the public set is poorly aligned with the private task. We present FedMKS (Federated Learning with Model and Knowledge Sharing), a hybrid algorithm that organises clients into architecture tiers and couples intra-tier weight averaging with cross-tier soft-label distillation, recovering the accuracy of weight sharing while preserving architectural diversity. We apply FedMKS to indoor Wi-Fi CSI sensing for occupancy and activity detection—a privacy-critical setting where raw data cannot leave the home—and evaluate it on the HomeOccupancy CSI dataset and MNIST/CIFAR-10 under IID and non-IID partitions against FedMD, FedAKD, FedAvg, FedProx, FedDF, and Local/Centralised baselines. FedMKS attains 90.4% on HomeOccupancy IID—within 1.3 percentage points of the centralised ceiling. Code and datasets are publicly available.

I. INTRODUCTION

Federated Learning (FL) allows a population of edge devices to collaboratively train a model while keeping raw data on device [1]. This privacy-preserving paradigm is appealing wherever data is sensitive, scarce, or expensive to centralise—smart homes, healthcare, and mobile sensing being canonical examples.

A persistent obstacle is *system heterogeneity*: edge devices range from constrained sensors to capable gateways, so no single architecture fits all. Weight-sharing methods (FedAvg, FedProx) require identical architectures and therefore force every client onto the smallest common model. Knowledge-distillation methods (FedMD, FedAKD [2], FedDF) remove this constraint by exchanging soft labels on a shared public set rather than weights, letting each client keep an architecture matched to its hardware.

Neither channel alone, however, is robust under *statistical heterogeneity*: when client label distributions are non-IID, weight averaging suffers client drift—on our low-data CSI benchmark it collapses to chance level on two of three seeds—while soft-label distillation provides a weaker per-round signal

whose quality depends on how well the public set matches the private task. This motivates a *hybrid* architecture that combines both channels—averaging weights where architectures match and distilling soft labels where they do not.

We instantiate this idea for indoor Wi-Fi sensing: Channel State Information (CSI) encodes fine-grained multipath propagation effects that shift when a person moves, breathes, or changes posture, enabling camera-free occupancy and activity recognition from commodity access points [3], [4]. Labelled CSI is private and hard to collect at scale, making it a natural application for heterogeneous FL.

Contribution. We propose **FedMKS** (Federated Learning with Model and Knowledge Sharing), which (i) partitions clients into architecture tiers and shares model weights within each tier, (ii) shares distilled soft labels across tiers on a public set, and (iii) applies Mixup augmentation before distillation. A dual-head tiered model decouples private classification from public-set distillation, so clients of different capacity collaborate without exchanging raw data.

II. RELATED WORK

Figure 2 sketches the FL research timeline relevant to this work; we review federated learning in general, distillation-based FL for system heterogeneity, the impact of statistical heterogeneity, and Wi-Fi sensing as our target application.

A. Federated Learning

FedAvg [1] introduced federated averaging of locally trained weights and remains the canonical FL baseline, but it requires identical architectures across clients; FedProx [6] adds a proximal term that tolerates statistical and system heterogeneity while keeping the same architectural constraint. Sozinov et al. [7] first showed federated HAR can rival centralised accuracy while preserving privacy; Wu et al. [8] extended FL to CSI sensing, where non-IID per-home distributions degrade FedAvg and motivate personalised approaches [9]–[11].

B. Knowledge-Distillation-Based FL

FedMD [12] removes the architecture constraint by sharing only soft predictions on a public set, building on Hinton’s distillation framework [13]; federated mutual learning [14] distils peer-to-peer. FedDF [5] instead ensembles client models

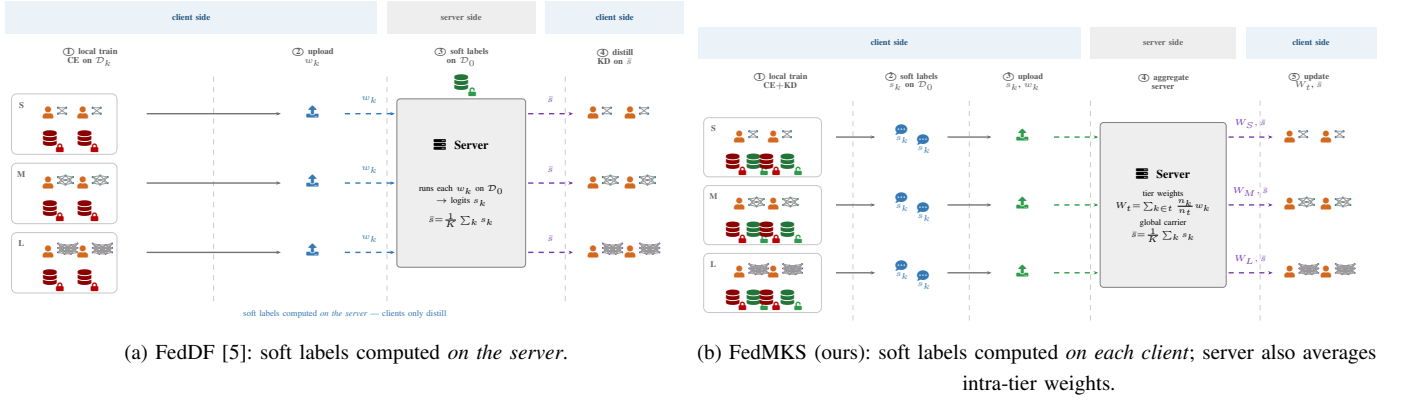


Fig. 1. **FedDF vs. FedMKS — one communication round.** FedDF uploads client weights and runs every model on the public set at the server; FedMKS computes soft labels s_k on each client, so the server only averages tensors and additionally FedAves intra-tier weights (W_S, W_M, W_L).

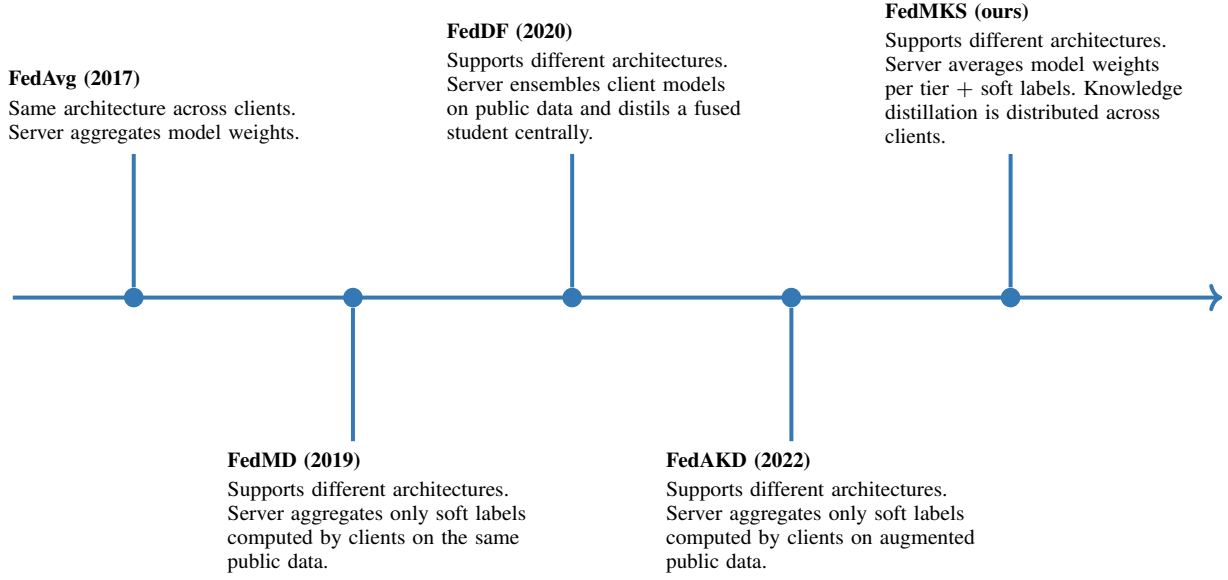


Fig. 2. Timeline of federated-learning research relevant to this work.

on the server and distills a fused student centrally. FedAKD [2] extends FedMD with Mixup-augmented public data for sensor-based HAR. Unlike FedDF, FedMKS distributes distillation across clients, removing the server bottleneck at the cost of higher client workload.

C. Statistical Heterogeneity

Zhao et al. [15] quantify the sharp accuracy drop of FL under non-IID data. Weight averaging suffers from client drift—on our low-data CSI benchmark it collapses to chance level in two of three seeds, though the effect is milder on data-rich image tasks; soft-label distillation avoids drift but provides a weaker per-round signal that degrades when the public set is cross-domain. Neither channel is uniformly reliable, motivating hybrid designs that combine both.

D. Wi-Fi Sensing

Wi-Fi CSI sensing has progressed from coarse localisation to fine-grained activity and gesture recognition [3], [4]. Early

systems showed that commodity 802.11 reflections encode human presence and activities [16], [17]; later work added real-time gesture classification [18], healthcare HAR [19], environment robustness [20], and joint localisation–recognition [21]. Indoor occupancy detection is a representative privacy-critical application: raw CSI reveals presence and behaviour inside a home and cannot be centralised, making it a natural fit for heterogeneous FL.

III. SYSTEM MODEL

A. Notation

B. Network and Data Model

Consider a star-topology FL network with a central server and $K=20$ clients. Each client k holds a private labelled dataset $\mathcal{D}_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k}$, where $x_i^k \in \mathbb{R}^{T \times (4+B)}$ is a preprocessed CSI window and $y_i^k \in \mathcal{Y}_k \subseteq \mathcal{Y}$ its activity label. Under *IID* partitioning each client receives $n_c=30$ examples per class; under *non-IID*, class proportions follow $\text{Dir}(\alpha=0.5)$. All clients

TABLE I
SYMBOL TABLE.

Symbol	Meaning
K	Number of clients ($K=20$)
$\mathcal{K}$	Client index set $\{1, \dots, K\}$
R	Number of FL communication rounds ($R=50$)
r	Round index, $r \in \{1, \dots, R\}$
$\mathcal{D}_k$	Private labelled dataset of client k
n_k	$ \mathcal{D}_k $; $n = \sum_k n_k$
$\mathcal{Y}$	Global label space; $C= \mathcal{Y} $
$\mathcal{Y}_k$	Local label subset, $\mathcal{Y}_k \subseteq \mathcal{Y}$
$\mathcal{D}^{\text{pub}}$	Shared public (unlabelled) dataset
t_k	Tier of client k : S, M, or L
θ_k	All trainable parameters of model f_k
θ_k^{bb}	Backbone parameters (shared by both heads)
$\mathbf{W}_t^{(r)}$	Server-side averaged model weights for tier t at round r
d	Distillation-head output dimension (soft-label size)
$s_k^{(r)}$	Soft-label matrix from client k at round r , $s_k^{(r)} \in \mathbb{R}^{ \mathcal{D}^{\text{pub}} \times d}$
$\bar{s}^{(r)}$	Aggregated global soft labels at round r
T, B	CSI window length (1500) and STFT bins (8)

also share an unlabelled *public* dataset $\mathcal{D}^{\text{pub}}$ (e.g. HomeHAR) used exclusively for knowledge transfer.

C. Heterogeneous Model Architecture

Each client is assigned a compute tier $t_k \in \{\text{S}, \text{M}, \text{L}\}$ reflecting its hardware budget. Every tier instantiates a *dual-head* model $f_k = (\phi_k, h_k^{\text{clf}}, h_k^{\text{dist}})$, where ϕ_k is a shared backbone parameterised by θ_k^{bb} and the two heads are trained with decoupled losses:

- **Classification head** h_k^{clf} : trained on private data with cross-entropy loss $\mathcal{L}_{\text{CE}}(\theta_k) = -\frac{1}{n_k} \sum_i \log p_k(y_i^k | x_i^k)$.
- **Distillation head** h_k^{dist} : produces a d -dimensional soft-label vector $s_k^{(r)} = h_k^{\text{dist}}(\phi_k(\mathcal{D}^{\text{pub}})) \in \mathbb{R}^{|\mathcal{D}^{\text{pub}}| \times d}$, aligned to the global aggregate via $\mathcal{L}_{\text{KD}}(\theta_k) = \frac{1}{|\mathcal{D}^{\text{pub}}|} \|s_k^{(r)} - \bar{s}^{(r-1)}\|_F^2$.

D. Traditional FL Objective and Its Limitations

FedAvg [1] seeks a single global model minimising the weighted empirical risk:

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_{\text{CE}}^k(\theta), \quad (1)$$

aggregating as $\theta^{(r)} \leftarrow \sum_k (n_k/n) \theta_k^{(r)}$. This requires *identical architectures* across clients; FedProx's proximal term $\frac{\mu}{2} \|\theta_k - \theta^{(r-1)}\|^2$ bounds client drift but keeps the same constraint.

E. Knowledge-Distillation-Based FL Objective

FedMD [12] instead aggregates shared representations: the server maintains global soft labels $\bar{s}^{(r)} = \frac{1}{K} \sum_k s_k^{(r)}$ and each client minimises

$$\mathcal{L}_k(\theta_k) = \underbrace{\mathcal{L}_{\text{CE}}(\theta_k; \mathcal{D}_k)}_{\text{private classification}} + \underbrace{\mathcal{L}_{\text{KD}}(\theta_k; \mathcal{D}^{\text{pub}}, \bar{s}^{(r-1)})}_{\text{KD alignment}}, \quad (2)$$

where the two terms are optimised in alternation: the classification head on $\mathcal{D}_k$ and the distillation head on $\mathcal{D}^{\text{pub}}$ (Section IV). Since $\bar{s}^{(r)}$ lives in a shared d -dimensional space, arbitrary backbones can join the same round.

TABLE II

PER-TIER BACKBONE ARCHITECTURES. EVERY TIER SHARES IDENTICAL INPUT AND OUTPUT SHAPES; ONLY THE BACKBONE DIFFERS.
CONV1D(f, k)/CONV2D(f, k): CONVOLUTION WITH f FILTERS, KERNEL SIZE k ; MP: MAXPOOL(2); GAP: *global average pooling* OVER THE TIME DIMENSION (AVERAGES THE SEQUENCE INTO A SINGLE VECTOR, AVOIDING A LARGE FLATTEN $\rightarrow$ DENSE WEIGHT MATRIX); BiLSTM(u): BIDIRECTIONAL LSTM WITH u UNITS PER DIRECTION; TB: TRANSFORMER ENCODER BLOCK (4-HEAD MHA, FFN-256, LAYER NORM); DO(p): DROPOUT(p).

Tier	Architecture
<i>CSI input</i> $x \in \mathbb{R}^{1500 \times 12}$ ($T=1500$, $4+B=12$ features)	
S	Conv1D(128,3) $\rightarrow$ MP $\rightarrow$ Conv1D(256,3) $\rightarrow$ MP $\rightarrow$ Conv1D(256,3) $\rightarrow$ GAP $\rightarrow$ Dense(256) $\rightarrow$ DO(0.30)
M	Conv1D(64,5) $\rightarrow$ MP $\rightarrow$ Conv1D(128,3) $\rightarrow$ MP $\rightarrow$ BiLSTM(128) $\rightarrow$ DO(0.30)
L	Conv1D(64,5) $\rightarrow$ MP $\rightarrow$ Conv1D(128,3) $\rightarrow$ MP $\rightarrow$ BiLSTM(128) $\rightarrow$ TB $\rightarrow$ GAP $\rightarrow$ DO(0.30)
<i>Image input</i> $x \in \mathbb{R}^{28 \times 28 \times 1}$ (MNIST) or $\mathbb{R}^{32 \times 32 \times 3}$ (CIFAR-10)	
S	Conv2D(32,3) $\rightarrow$ MP $\rightarrow$ Flatten $\rightarrow$ Dense(128) $\rightarrow$ DO(0.25)
M	Conv2D(64,3) $\rightarrow$ MP $\rightarrow$ Conv2D(128,3) $\rightarrow$ MP $\rightarrow$ Flatten $\rightarrow$ Dense(256) $\rightarrow$ DO(0.30)
L	Conv2D(64,3) $\rightarrow$ MP $\rightarrow$ Conv2D(128,3) $\rightarrow$ MP $\rightarrow$ Conv2D(256,3) $\rightarrow$ MP $\rightarrow$ Flatten $\rightarrow$ Dense(512) $\rightarrow$ DO(0.40)
classification head: Dense(C)+softmax; distillation head: Dense($d=64$)+ReLU	

F. Problem Definition

Goal. Given K clients with heterogeneous private datasets $\{\mathcal{D}_k\}$ and heterogeneous model tiers $\{t_k\}$, find per-client classifiers $\{f_k^{\text{clf}}\}$ that jointly maximise mean validation accuracy across clients, subject to two constraints:

- 1) *Privacy*: $\mathcal{D}_k$ never leaves client k ; only model parameters and/or soft labels are transmitted.
- 2) *Architectural heterogeneity*: clients in different tiers ($t_j \neq t_k$) instantiate different architectures, so their weight spaces are incompatible and cannot be aggregated directly; cross-tier knowledge transfer must occur in a shared representation space rather than in weight space.

FedMKS (Section IV) meets both constraints by combining per-tier weight averaging (Eq. 1) with cross-tier soft-label distillation (Eq. 2).

IV. METHODS

A. Tiered Dual-Head Architecture

The three tiers use distinct backbones of increasing capacity (Table II). All tiers consume the *same* input shape and expose the *same* two output heads; only the backbone differs. The *classification head* (Dense $\rightarrow$ Softmax, CE loss) is trained on $\mathcal{D}_k$; the *distillation head* (Dense $\rightarrow$ ReLU with $d=64$ output units, MSE loss on $\bar{s}$) is trained on $\mathcal{D}^{\text{pub}}$. Gradient flows are fully decoupled between heads; only the backbone is shared.

B. FedMKS: Model and Knowledge Sharing

FedMKS (Fig. 1) addresses the limitations of both FedAvg and FedMD: FedAvg forces every client onto the smallest shared architecture, while FedMD shares only soft labels and forfeits the accuracy gains of weight sharing. FedMKS simultaneously (i) FedAvg's weights within each tier, (ii) distills knowledge across tiers via aggregated soft labels on the public

Algorithm 1 FedMKS — one communication round

Require: Clients $\mathcal{K}$; tiers $t \in \{\text{S, M, L}\}$; $\mathcal{D}^{\text{pub}}$; E_{kd} , E_{loc} ; round r ; current global soft labels $\bar{s}^{(r-1)}$ and tier weights $\{\mathbf{W}_t^{(r-1)}\}$

Ensure: Updated global soft labels $\bar{s}^{(r)}$ and tier weights $\{\mathbf{W}_t^{(r)}\}$

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1: if  $r = 1$  then
2:   Server: send empty payload to each  $k$  ▷ bootstrap
3: else
4:   Server: broadcast  $\bar{s}^{(r-1)}$  and  $\mathbf{W}_{t_k}^{(r-1)}$  to each  $k$ 
5: end if
6: for each client  $k$  in parallel do
7:   if  $r > 1$  and  $\mathbf{W}_{t_k}^{(r-1)}$  received then
8:     Load model weights  $\leftarrow \mathbf{W}_{t_k}^{(r-1)}$ 
9:     Receive soft labels  $\bar{s}^{(r-1)}$  as public-set targets
10:    Fine-tune dist. head on  $\mathcal{D}^{\text{pub}}$  for  $E_{\text{kd}}$  epochs minimising
         $\mathcal{L}_{\text{MSE}}(f_k^{\text{dist}}(\cdot), \bar{s}^{(r-1)})$ 
11:   end if
12:   Train clf. head on private data  $\mathcal{D}_k$  for  $E_{\text{loc}}$  epochs minimising
         $\mathcal{L}_{\text{CE}}$ 
13:   Return  $s_k^{(r)} \leftarrow f_k^{\text{dist}}(\mathcal{D}^{\text{pub}})$  and  $w_k^{(r)} \leftarrow \theta_k$ 
14: end for
15: Server:  $\bar{s}^{(r)} \leftarrow \frac{1}{K} \sum_{k=1}^K s_k^{(r)}$ 
16: Server:  $\mathbf{W}_t^{(r)} \leftarrow \text{FedAvg}(\{w_k^{(r)} : t_k=t\})$  for each tier  $t$ 

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TABLE III
ALGORITHM SUMMARY

Algo.	Soft labels	Wts	Topology
FedMD	✓	–	Heterogeneous
FedAKD	✓	–	Heterogeneous
FedMKS	✓	✓	Semi-het.
FedDF	✓	✓	Heterogeneous
FedAvg	–	✓	Homogeneous
FedProx	–	✓	Homogeneous
Local	–	–	Isolated
Central	–	–	Centralised

set, and (iii) applies Mixup augmentation [22] to the public set (following FedAKD [2]).

In round $r=1$ each client trains only on private data and uploads its initial soft labels and model weights. From $r=2$ the server returns $\mathbf{W}_{t_k}^{(r-1)}$ and $\bar{s}^{(r-1)}$; the client loads the weights, fine-tunes the distillation head on $\bar{s}^{(r-1)}$ targets, trains the classification head on $\mathcal{D}_k$ for E_{loc} epochs, and returns $s_k^{(r)}$ and $w_k^{(r)}$.

Algorithm 1 formalises one FedMKS round.

Comparison with FedDF. FedDF [5] runs all client models on $\mathcal{D}^{\text{pub}}$ at the server, forms an ensemble $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}; \theta_k)$, and trains a fused student by minimising $D_{\text{KL}}(\bar{s}_\tau \| f_\tau(\mathcal{D}^{\text{pub}}; \theta))$, incurring $O(KP\mathcal{F} + E_{\text{kd}}P(\mathcal{F} + \mathcal{B}))$ operations per round ($P=|\mathcal{D}^{\text{pub}}|$; $\mathcal{F}$, $\mathcal{B}$ = forward/backward cost). FedMKS inverts this: the server only averages tensors, $O(\sum_t |\mathbf{W}_t| + KPd)$, shifting distillation to clients—far better suited to a lightweight edge coordinator.

Table III summarises all evaluated algorithms.

TABLE IV
HYPERPARAMETER SETTINGS (ALL EXPERIMENTS).

Parameter	Value
Rounds R	50
Clients K	20
IID samples / class / client	30 (CSI) / 100 (image)
Non-IID Dirichlet α	0.5
Local epochs E_{loc}	4 (CSI) / 5 (image)
KD epochs E_{kd}	2 (CSI) / 3 (image)
Soft-label dimension d	64
FedProx μ	0.01
Public alignment samples $ \mathcal{D}^{\text{pub}} $	1 375 (CSI) / 10 000 (image)
CSI window T	1500
STFT bins B	8

V. EXPERIMENTS AND RESULTS

A. Experiment Design

Datasets. *HomeOccupancy*: 3-class CSI (empty/sleep/work), sessions 1–2 as train, session 3 as test; public data is the *HomeHAR* CSI corpus. *MNIST*: 10-class digit images; public data is CIFAR-10. *CIFAR-10*: 10-class object images; public data is MNIST.

Hyperparameters (Table IV): image benchmarks use $n_c=100$, $E_{\text{loc}}=5$, $E_{\text{kd}}=3$; all other settings are shared. Results are mean validation accuracy over completed seeds (up to three); shaded bands in the figures show the seed range. Reported accuracy is the mean over clients of each client’s validation accuracy *after* its local update on the shared test set—the only metric defined uniformly across all methods, since the KD-based algorithms maintain no single global model. For FedAvg/FedProx this evaluates the deployed model after one round of local adaptation rather than the server-side aggregate.

The *main comparison* evaluates all 7 algorithms on 3 datasets under uniform tier heterogeneity (35% S/35% M/30% L), IID and non-IID. FedAvg and FedProx degrade every client to the small model; KD methods and FedMKS keep their assigned tiers. Two ablation studies follow: the first ablates the tier distribution for FedMD on HomeOccupancy (all_small, uniform, skewed=60/30/10); the second compares FedMKS and FedDF on HomeOccupancy (IID, single seed) on accuracy, server/client compute, and communication bytes.

B. Results

1) *Algorithm Comparison.* Figs. 3 and 4 show validation accuracy over 50 rounds under IID and non-IID partitions.

HomeOccupancy. Under IID, FedMKS (90.4%) and FedAvg (90.3%) approach the centralised ceiling (91.7%), while FedMD (88.5%) and FedAKD (87.0%) lag without weight sharing; Local plateaus at 81.4%. Under non-IID, the weight-sharing methods (FedAvg 47.2%, FedProx 43.2%) trail local-only training (60.1%), whereas the soft-label channel sustains 63.2% (FedMD) and 61.0% (FedAKD)—at or above local training—and the hybrid FedMKS reaches 53.8%, still above both weight-sharing methods. The gap is not graceful degradation: in two

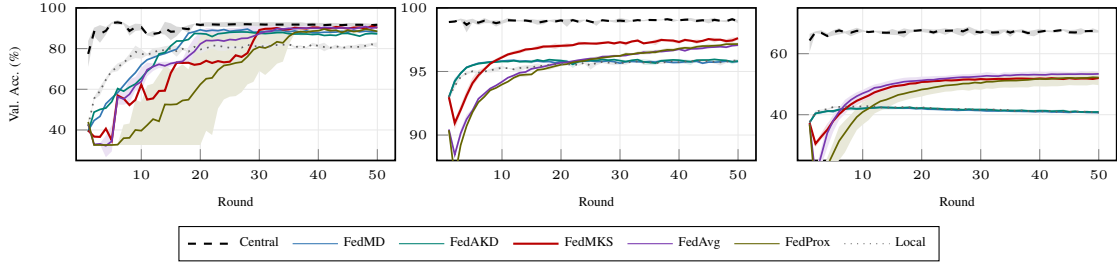


Fig. 3. Validation accuracy over 50 rounds under IID partitioning. Panels: HomeOccupancy, MNIST, CIFAR-10. Curves show the mean over completed seeds (up to three); shaded bands show the seed range. FedAvg/FedProx use the all-small architecture.

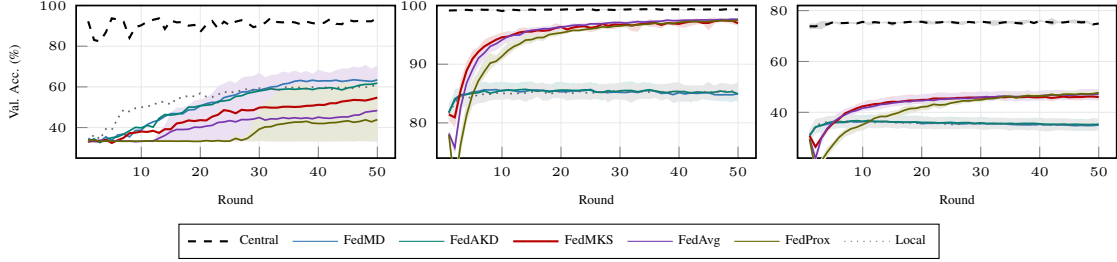


Fig. 4. Validation accuracy over 50 rounds under non-IID partitioning (Dirichlet $\alpha=0.5$). Panels: HomeOccupancy, MNIST, CIFAR-10. Curves show the mean over completed seeds; shaded bands show the seed range. FedAvg/FedProx use the all-small architecture.

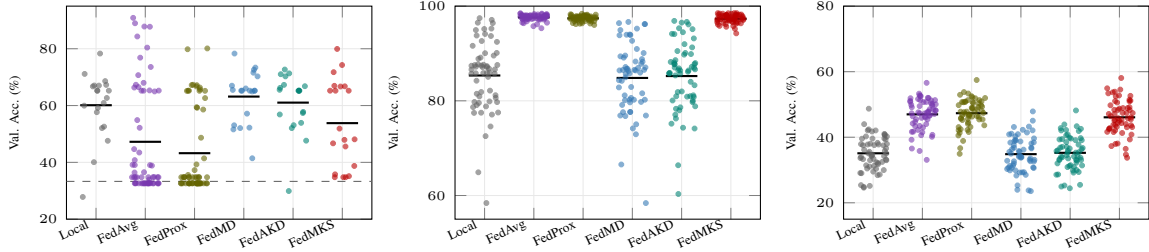


Fig. 5. Per-client final validation accuracy (mean of the last five rounds, evaluated on each client’s model *after* its local update) under non-IID partitioning. Each dot is one client; all completed seeds are pooled (60 dots per three-seed method, 20 per single-seed method). Black dashes mark method means; the grey dashed line marks chance level (33.3% for HomeOccupancy, 10% for CIFAR-10). Left: HomeOccupancy—in the two collapsed FedAvg/FedProx seeds *all* clients sit at chance, while KD-based and local clients spread around $\approx 60\%$. Middle/right: MNIST and CIFAR-10—no client collapses; weight sharing lifts every client above the local mean.

of three seeds the averaged model converges to a single-class predictor (validation accuracy $\approx 33\%$, the test majority share) within the first round and never recovers. With only ≈ 90 samples per client, most local datasets are dominated by one class, so each round’s local optimum is a majority-class predictor; averaging twenty incompatible single-class models keeps the global model in that basin, and skewed local updates cannot escape it. FedProx’s proximal term does not help—it anchors clients to the already-collapsed global model.

a) Per-client view.: Fig. 5 resolves this asymmetry at client granularity. Local-only clients are heterogeneous—most reach 50–78% while a few sit near chance—and average 60.1%. This spread is a partition effect, not an architecture effect (all local clients share the small-tier model): replaying the seeded Dirichlet draw shows the best client (78.3%) holds a mid-sized two-class partition (103 samples, 54% dominant-class share), while the worst (27.8%) has only 21 samples, 67%

dominated by one class; dominant-class share correlates with local accuracy at $r=-0.36$. The KD-based methods’ *global* accuracy matches that local mean almost exactly (FedMD 63.2%, FedAKD 61.0%), and the per-client spread is preserved as well: each client keeps its own model, so the shared soft labels act as a regulariser rather than replacing the local solution. Weight sharing behaves qualitatively differently because every client evaluates the *same* averaged model: in the two failed seeds all twenty clients sit at $\approx 33\%$ together, while in the surviving seed clients average 70.0% after their local updates and only two fall below 40%. Weight averaging therefore does not interpolate between strong and weak local models; it produces a single global model whose fate is sealed by whether the first-round average escapes the majority-class basin, which in turn depends on how many clients’ local minima are dominated by compatible classes. FedMKS lies between the extremes: its small-tier cluster is dragged to 35–48% by intra-

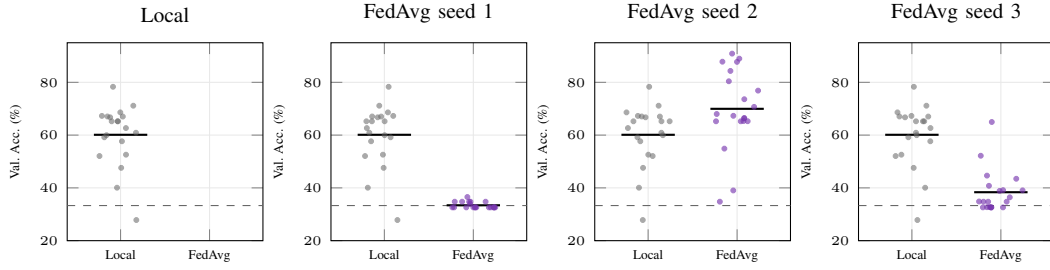


Fig. 6. HomeOccupancy non-IID, per-client final accuracy by seed (each client’s model evaluated after its local update). Grey dots (identical in every panel) are the Local-only clients; purple dots are FedAvg clients in seeds 1–3. In seeds 1 and 3 *every* FedAvg client sits at chance (33.3%)—the shared model collapses collectively—while seed 2 retains a healthy spread comparable to Local.

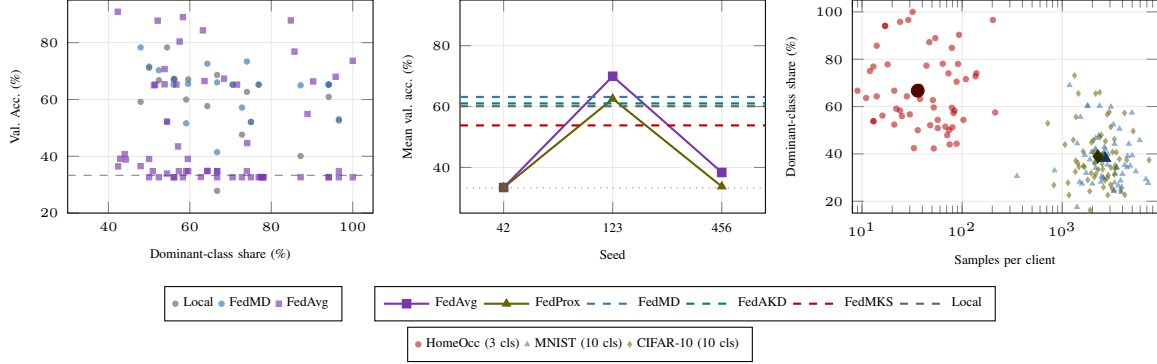


Fig. 7. Data distribution decides whether weight averaging survives. (a) HomeOccupancy non-IID: per-client final accuracy vs. each client’s dominant-class share. Local and FedMD degrade as the partition approaches a single class ($r=-0.36$ for Local); FedAvg clients in the two collapsed seeds sit at chance at *every* share—collapse is a property of the averaged model, not of any client’s data. (b) Mean accuracy per seed: FedAvg and FedProx collapse in two of three seeds while the KD-based methods and Local (single seed, dashed references) hold ≈ 54 – 63% ; FedMKS never falls below 53% , so its worst case exceeds the weight-sharing worst case by ≈ 20 points. (c) Per-client partitions of all benchmarks in the (samples, dominant-share) plane (all seeds pooled; large marks are dataset means). HomeOccupancy alone occupies the high-skew/low-data corner.

TABLE V

FINAL VALIDATION ACCURACY (%): MEAN OF THE LAST FIVE ROUNDS, AVERAGED OVER COMPLETED SEEDS ($\pm$ STD. DEV. WHERE MORE THAN ONE SEED COMPLETED). “—” MARKS SETTINGS WITHOUT A COMPLETED RUN. CORRESPONDS TO FIGS. 3 AND 4.

Algorithm	IID			non-IID		
	HomeOcc.	MNIST	CIFAR-10	HomeOcc.	MNIST	CIFAR-10
Central	91.7 $\pm$ 0.8	99.0 $\pm$ 0.0	67.2 $\pm$ 0.5	92.6	99.3 $\pm$ 0.0	75.2 $\pm$ 0.1
FedMD	88.5	95.8 $\pm$ 0.0	40.8 $\pm$ 0.3	63.2	84.8 $\pm$ 0.9	34.8 $\pm$ 1.9
FedAKD	87.0	95.9 $\pm$ 0.1	41.0 $\pm$ 0.3	61.0	85.2 $\pm$ 1.0	35.2 $\pm$ 1.7
FedMKS	90.4	97.5 $\pm$ 0.1	51.9 $\pm$ 0.4	53.8	97.3 $\pm$ 0.3	46.1 $\pm$ 1.5
FedAvg	90.3 $\pm$ 0.6	97.0 $\pm$ 0.1	53.3 $\pm$ 0.9	47.2 $\pm$ 16.2	97.6 $\pm$ 0.2	47.0 $\pm$ 1.6
FedProx	88.8 $\pm$ 0.5	97.1 $\pm$ 0.1	52.1 $\pm$ 1.8	43.2 $\pm$ 13.7	97.4 $\pm$ 0.2	47.3 $\pm$ 0.8
Local	81.4 $\pm$ 1.0	95.8 $\pm$ 0.1	40.9 $\pm$ 0.2	60.1	85.4 $\pm$ 1.0	35.1 $\pm$ 1.7

tier averaging while the medium and large tiers remain healthy, yielding 53.8%. On MNIST, where every local client exceeds 58%, the same mechanism is harmless—the shared model (97.6%) lifts *all* clients above the local mean (85.4%). Client drift is thus fatal only when local models are both weak and mutually incompatible, which is precisely the regime where the distillation channel earns its keep. Fig. 6 shows the same effect per seed: in the two failed seeds every FedAvg client sits at chance, while the surviving seed mirrors the Local spread.

MNIST. IID: Central 99.0%, FedMKS 97.5%, FedAvg/FedProx 97.0%/97.1%, FedMD/FedAKD/Local $\approx 95.8\%$.

Non-IID: FedAvg 97.6%, FedProx 97.4%, FedMKS 97.3%, while FedMD/FedAKD/Local plateau at $\approx 85\%$ — ≈ 12 points below the all-small weight-sharing methods despite their larger architectures.

CIFAR-10. Cross-domain public data (MNIST) gives little signal: all FL methods fall 10–30% below centralised. FedAvg (53.3%) edges FedProx (52.1%) and FedMKS (51.9%) under IID; under non-IID the three are close (FedProx 47.3%, FedAvg 47.0%, FedMKS 46.1%). KD-only methods (FedMD 34.8%, FedAKD 35.2%) remain at Local level (35.1%).

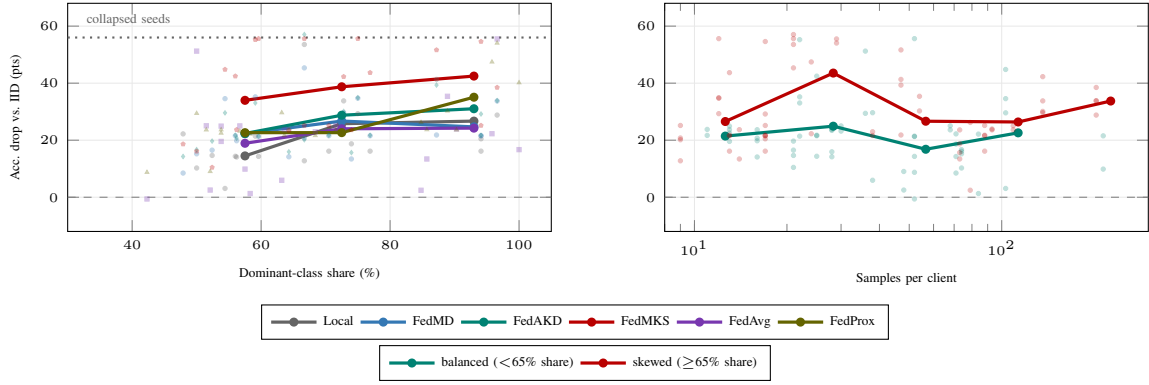


Fig. 8. Partition imbalance, not partition size, drives the per-client accuracy decline under non-IID (HomeOccupancy). Dots are individual clients; thick lines are binned means. The y-axis is the drop from the method’s own IID mean. (a) The decline grows steadily with the dominant-class share for every method that keeps per-client models—e.g. FedAvg in its surviving seed falls from ≈ 0 to ≈ 24 points of drop across the share range ($r=+0.73$ for FedProx)—while the collapsed FedAvg/FedProx seeds form a flat band at ≈ 56 points (dotted); their bias is total and independent of the partition. (b) The same decline vs. samples per client, pooled over methods and split by partition balance. More data helps only when the partition is balanced (teal line declines); for skewed clients extra samples are mostly more of the dominant class and the drop stays high at every size.

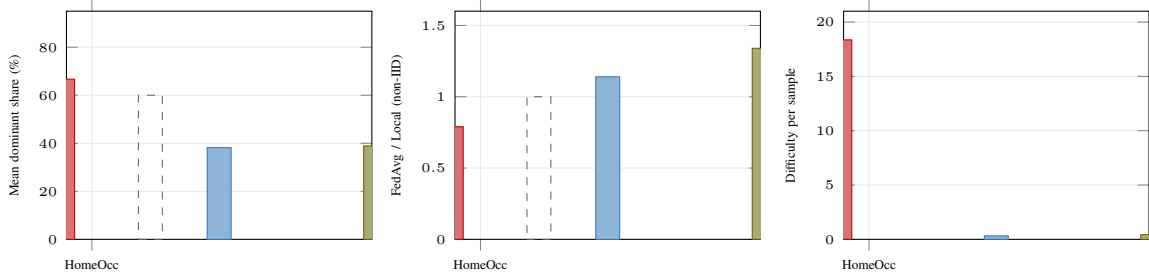


Fig. 9. Why weight averaging fails on HomeOccupancy. (a) Mean dominant-class share per client under $\text{Dir}(\alpha=0.5)$: only HomeOccupancy crosses the $\approx 60\%$ level at which a single class dominates the local optimum. (b) FedAvg-to-Local accuracy ratio under non-IID: sharing beats local training on MNIST and CIFAR-10 but loses $\approx 21\%$ on HomeOccupancy. (c) Difficulty per sample, $(1-\text{silhouette})/n_{\text{client}} (\times 10^3)$: combining low input-space separability with the smallest per-client partitions makes HomeOccupancy $\approx 45\times$ harder per sample than the image benchmarks.

b) Key observation.: Under non-IID, the better channel depends on the public–private domain match: on HomeOccupancy, soft-label distillation (FedMD 63.2%, FedAKD 61.0%) beats smallest-model weight sharing (47.2%), while on the image benchmarks weight sharing dominates (MNIST: $\approx 97\%$ vs. $\approx 85\%$). FedMKS (53.8%) sits between the two channels on HomeOccupancy—its intra-tier averaging inherits part of the drift penalty—but remains the only method that is competitive on both sides of the domain-match divide.

c) What decides performance?: Three candidate factors emerge from our results, and they are not equal. (i) *Seed randomness*: under non-IID the partition draw alone decides whether weight averaging retains or destroys local structure—two of three seeds collapse, one survives (Figs. 6 and 7(b)). (ii) *Model capacity*: within every tier distribution the per-tier means differ by only a few points (Fig. 10), so capacity is a second-order factor under soft-label sharing. (iii) *Data distribution*: the dominant factor. Fig. 7(a) ties each client’s accuracy to the dominant-class share of its partition ($r=-0.36$ for Local), and Fig. 7(c) shows why the benchmarks behave so differently: HomeOccupancy clients sit in the high-skew/low-data corner (≈ 52 samples, $\approx 67\%$ dominant share on average), where a single class can dominate both the local optimum and

the first global average, while MNIST/CIFAR-10 clients hold $\approx 60\times$ more samples at lower shares. Fig. 9 quantifies the gap: HomeOccupancy is the only benchmark whose mean dominant share crosses the $\approx 60\%$ single-class regime, the only one where weight averaging loses to local training (FedAvg/Local $=0.79$), and—combining low input-space separability with the smallest partitions— $\approx 45\times$ harder per sample than the image benchmarks. Fig. 8 makes the imbalance–bias link explicit: the per-client accuracy decline relative to IID grows with the dominant-class share for every method that keeps per-client models ($r=+0.36$ for Local, $r=+0.73$ for FedProx in its surviving seed). Raw sample count alone does not predict the decline ($r\approx 0$), but for a subtler reason than one might expect: stratifying by balance shows that additional data reduces the drop only for balanced clients—for skewed clients the extra samples are mostly more of the dominant class and the drop stays high at every size (Fig. 8(b)). How much data, and how balanced, is *enough* for a given task difficulty is the natural next question (Sec. VII).

Crucially, this is also where FedMKS’s robustness shows: in the two collapsed seeds every FedAvg/FedProx client sits at chance, while FedMKS’s worst case (53.8%) exceeds the weight-sharing worst case (33.4%) by ≈ 20 points (Fig. 7(b)).

TABLE VI

FEDMD TIER-HETEROGENEITY ABLATION ON HOME OCCUPANCY: FINAL VALIDATION ACCURACY (%), MEAN OF THE LAST FIVE ROUNDS OVER COMPLETED SEEDS. CORRESPONDS TO FIG. 10.

Tier distribution	IID	non-IID
all_small	86.7±0.8	59.0±2.0
uniform	86.7±0.5	61.5±0.7
skewed	85.8±1.2	58.8

Robustness to heterogeneity therefore holds on both axes named in our title: *system* heterogeneity, through tiered partial model sharing, and *statistical* heterogeneity, where the distillation channel bounds the damage that weight averaging can do.

2) Ablation Studies:

a) *Tier heterogeneity.*: Fig. 10 ablates the tier distribution for FedMD on HomeOccupancy. Under IID all configurations converge to $\approx 86\%$; under non-IID *uniform* (61.5%) beats *all_small* (59.0%) and *skewed* (58.8%)—small-dominated tiers underperform a balanced mix under label skew. The client-level panels show why the gap is small: within every configuration the tier means differ by only a few points, so per-client accuracy under soft-label sharing is governed by the data partition rather than model capacity.

b) *FedMKS vs. FedDF.*: Fig. 11 and Table VII compare FedMKS and FedDF on HomeOccupancy (IID, single seed, 30 rounds, 10 clients). FedDF converges faster (plateau near round 10, 89.0% vs. 87.1% final-five mean), though FedMKS is still climbing and edges ahead at round 30 (88.2% vs. 88.1%). The efficiency profiles differ sharply: FedDF spends ≈ 243 s/round on server-side ensemble training, whereas FedMKS does essentially no server compute (≈ 0.06 s) and distributes distillation across parallel clients (≈ 93 s vs. ≈ 25 s per client). FedMKS removes the serial server bottleneck at $\approx 22\%$ higher communication (37.9 vs. 31.0 MB/round); the whole ≈ 6.9 MB overhead is the soft-label exchange— $|\mathcal{D}^{\text{pub}}| \times d = 1375 \times 64$ floats (≈ 0.34 MB) uploaded and broadcast per client per round—which FedDF avoids by distilling on the server. The dashed curve in Fig. 11 shows FedMKS with Mixup disabled on the public set: convergence is comparable and the final accuracy drops only ≈ 1 pp (87.3% vs. 88.2% at round 30), indicating that Mixup yields a modest regularisation benefit rather than being essential to the hybrid update.

VI. CONCLUSION

We presented **FedMKS**, a heterogeneous federated knowledge-distillation algorithm for CSI-based indoor sensing that couples intra-tier weight sharing with cross-tier soft-label distillation. FedMKS reaches near-centralised accuracy on HomeOccupancy ($\Delta \approx 1.3$ points under IID) while preserving architectural heterogeneity, and approaches FedDF accuracy (87.1% vs. 89.0%) while eliminating its ≈ 243 s/round server bottleneck (≈ 0.06 s/round) at $\approx 22\%$ higher communication. FedMKS allows heterogeneity; it exceeds model sharing when model sharing collapses (HomeOccupancy non-IID), and

TABLE VII

EFFICIENCY COMPARISON ON HOME OCCUPANCY (PER ROUND UNLESS NOTED; ACCURACY = FINAL-FIVE-ROUND MEAN).

Metric (per round)	FedMKS	FedDF
Final val. accuracy (%)	87.1	89.0
Server compute (s)	0.06	243.2
Mean client compute (s)	93.1	24.6
Comm.: model weights (MB)	31.0	31.0
Comm.: soft labels (MB)	6.9	0.0
Communication total (MB)	37.9	31.0
Total communication (MB, 30 r.)	1137.7	930.1

it follows model sharing when model sharing significantly outperforms knowledge distillation (MNIST, CIFAR-10). Its robustness therefore covers both axes of heterogeneity: *system* heterogeneity, via tiered partial model sharing that pure weight-sharing methods cannot support, and *statistical* heterogeneity, where its worst case (53.8%) stays ≈ 20 points above the weight-sharing worst case (33.4%) under label skew.

VII. LIMITATIONS AND FUTURE WORK

In this study, we did not characterise the impact of public-private domain alignment on soft-label distillation. As Fig. 3 shows, the soft-label-only algorithms (FedMD, FedAKD) remain competitive on HomeOccupancy, where the public HomeHAR pool shares the same CSI sensing modality, but degrade to near-local performance on the image benchmarks, where the public set is cross-domain (MNIST $\leftrightarrow$ CIFAR-10). A systematic evaluation of KD-based algorithms with public datasets whose characteristics match those of the private task is needed to quantify this effect. Finally, our results suggest that the effectiveness of federated averaging is governed by the ratio between task difficulty and the quantity and distribution of per-client data (Fig. 7(b)); formalising this into a predictive difficulty metric—e.g. the minimum per-client sample count and maximum tolerable dominant-class share for a task of a given difficulty—is a promising direction for a future extension of this work.

CODE AND DATA AVAILABILITY

Source code is available at <https://github.com/gadm21/HomeSenseKD>. HomeOccupancy and HomeHAR datasets are hosted on HuggingFace at <https://huggingface.co/gadgadgad>.

REFERENCES

- [1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. AISTATS*, Fort Lauderdale, FL, 2017, pp. 1273–1282.
- [2] G. Gad and Z. M. Fadlullah, “Federated learning via augmented knowledge distillation for heterogeneous deep human activity recognition systems,” *Sensors*, vol. 23, no. 1, p. 6, 2023.
- [3] Y. Ma, G. Zhou, and S. Wang, “Wifi sensing with channel state information: A survey,” *ACM Comput. Surv.*, vol. 52, no. 3, pp. 1–36, 2019.
- [4] S. Yousefi, H. Narui, S. Dayal, S. Ermon, and S. Valaei, “A survey on behavior recognition using WiFi channel state information,” *IEEE Commun. Mag.*, vol. 55, no. 10, pp. 98–104, 2017.

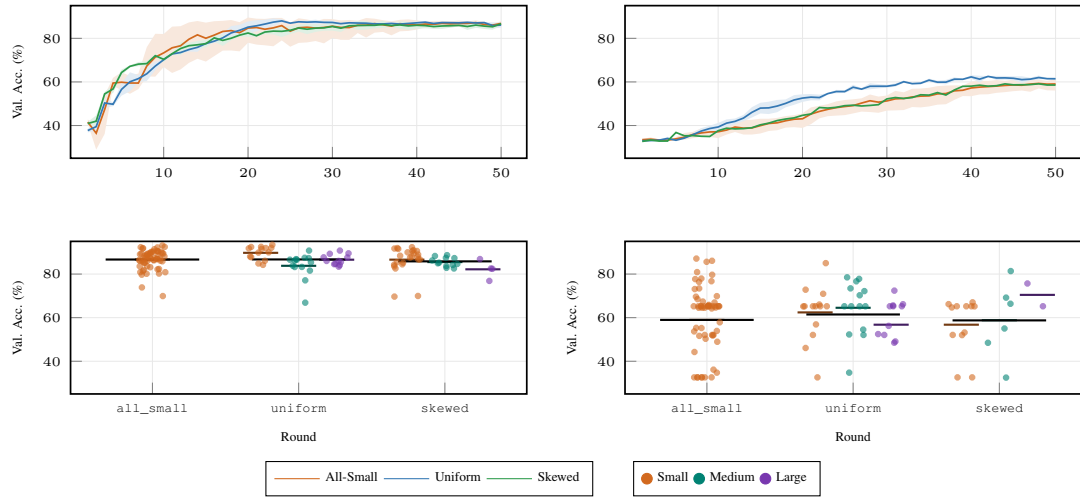


Fig. 10. FedMD tier-heterogeneity ablation on HomeOccupancy. Top: mean validation-accuracy curves over completed seeds (shaded bands: seed range), IID (left) and non-IID (right) on a shared y-axis. Bottom: per-client final accuracy (mean of the last five rounds, seeds pooled). Each distribution column is subdivided by architecture tier—small (orange), medium (teal), large (purple)—so tier counts are visible (e.g. skewed: 12/6/2); short coloured dashes mark tier means, the long black dash the column mean. Tier means differ by only a few points within every setting: under soft-label sharing, per-client accuracy is governed by the data partition, not by model capacity.

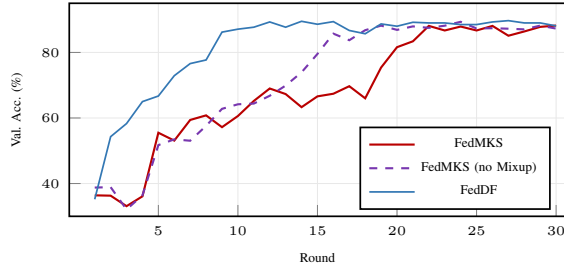


Fig. 11. FedMKS vs. FedDF validation accuracy on HomeOccupancy (IID, 30 rounds, 10 clients). The dashed curve is a FedMKS ablation with Mixup augmentation on the public set disabled.

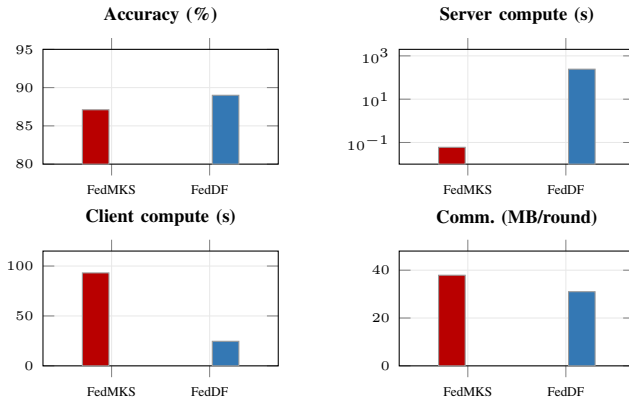


Fig. 12. Compute, communication, and accuracy comparison of FedMKS and FedDF on HomeOccupancy (cf. Table VII). FedMKS eliminates the server bottleneck at $\approx 22\%$ higher communication and higher per-client compute.

- [5] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, “Ensemble distillation for robust model fusion in federated learning,” in *Proc. NeurIPS*, 2020, pp. 2351–2363.
- [6] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith,

- “Federated optimization in heterogeneous networks,” in *Proc. MLSys*, 2020, pp. 429–450.
- [7] K. Sozinov, V. Vlassov, and S. Girdzijauskas, “Human activity recognition using federated machine learning,” in *Proc. IEEE AICCSA*, Aqaba, Jordan, 2018, pp. 1–6.
- [8] Q. Wu, K. He, and X. Chen, “Personalized federated learning for intelligent IoT applications: A cloud-edge based framework,” *IEEE Open J. Comput. Soc.*, vol. 1, pp. 35–44, 2021.
- [9] S. Ek, F. Portet, P. Lalanda, and G. Vega, “Evaluation of federated learning aggregation algorithms: Application to human activity recognition,” in *Proc. IoT*, 2020, pp. 1–7.
- [10] X. Ouyang, Z. Shuai, J. Zhou, I. W. Shi, Z. Xie, G. Xing, and J. Huang, “ClusterFL: A similarity-aware federated learning system for human activity recognition,” in *Proc. ACM MobiSys*, 2021, pp. 54–66.
- [11] R. Presotto, G. Civitarese, and C. Bettini, “SemFed: Semantic-based federated learning for heterogeneous activity recognition,” in *Proc. IEEE PerCom*, 2022, pp. 1–10.
- [12] D. Li and J. Wang, “FedMD: Heterogeneous federated learning via model distillation,” in *NeurIPS Workshop on Federated Learning*, Vancouver, 2019.
- [13] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” in *NeurIPS Deep Learning Workshop*, Montreal, 2015.
- [14] T. Shen, J. Zhang, X. Jia, F. Zhang, G. Huang, P. Zhou, K. Cheng, and F. Wu, “Federated mutual learning,” arXiv preprint arXiv:2006.16765, 2020.
- [15] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, “Federated learning with non-IID data,” arXiv preprint arXiv:1806.00582, 2018.
- [16] F. Adib and D. Katabi, “See through walls with WiFi!” in *Proc. ACM SIGCOMM*, Hong Kong, 2013, pp. 75–86.
- [17] D. Wang, D. Bhatt, A. Sheth, B. Krishnamachari, V. Srinivasan, and J. Cai, “E-eyes: Device-free location-oriented activity identification using fine-grained WiFi signatures,” in *Proc. ACM MobiCom*, Maui, HI, 2014, pp. 617–628.
- [18] Z. Chen, L. Zhang, C. Jiang, Z. Cao, and W. Cui, “WiFi CSI based passive human activity recognition using attention based BLSTM,” *IEEE Trans. Mobile Comput.*, vol. 18, no. 11, pp. 2714–2724, 2018.
- [19] C. Han, K. Bhanu, R. Srivastava, and L. Zhang, “WiFi-based activity recognition for healthcare using deep learning,” *IEEE Access*, vol. 8, pp. 56–68, 2020.
- [20] C. Jiang, J. Zhao, X. Chen, and H. Liu, “Towards environment-independent device-free human activity recognition,” in *Proc. ACM MobiCom*, New Orleans, LA, 2021, pp. 289–302.
- [21] X. Liu, J. Cao, S. Tang, J. Wen, and P. Guo, “Contactless respiration monitoring via off-the-shelf WiFi devices,” *IEEE Trans. Mobile Comput.*, vol. 15, no. 10, pp. 2466–2479, 2020.

- [22] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, “mixup: Beyond empirical risk minimization,” in *Proc. ICLR*, 2018.